21ECC301P - MICROPROCESSOR, MICROCONTROLLER AND INTERFACING TECHNIQUES

SEMESTER V

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IoT based Patient Monitoring System

A PROJECT REPORT

Submitted by

Dharsana S [Reg No: RA2311053010030]

Selvi Harini A R [Reg No: RA2311053010031]

Madhumitha V [Reg No: RA2311053010057]

Nithin Raj T [Reg No: RA2311053010060]

Under the guidance of

Dr. Diwakar R Marur

(Associate Professor, Department of Electronics & Communication Engineering)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

College of Engineering and Technology

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

S.R.M. Nagar, Kattankulathur – 603203, Chengalpattu District.

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COLLEGE OF ENGINEERING & TECHNOLOGY
SRM INSTITUTE OF SCIENCE & TECHNOLOGY
S.R.M. Nagar, Kattankulathur – 603203, Chengalpattu District

BONAFIDE CERTIFICATE

Certified that this project report titled "IoT Based Patient Monitering System" is the bonafide work of "DHARSANA S [RA2311053010030], SELVI HARINI A R [RA2311053010031], MADHUMITHA V [RA2311053010057] and NITHIN RAJ T [RA2311053010060]", who carried out the project work under my supervision, as a part of the course **"Microprocessor, Microcontroller and Interfacing Techniques (21ECC301P)",** during the academic year 2025-26 (ODD) in the Department of ECE, SRM Institute of Science & Technology, Kattankulathur.

SUPERVISOR	COURSE IN CHARGE
Dr. Diwakar R Marur	Dr. Diwakar R Marur
Associate Professor	Associate Professor
Dept. of Electronics and	Dept. of Electronics and
Communication Engineering	Communication Engineering

Date: **Academic Advisor**

(Dr. Diwakar R Marur)

DECLARATION

We hereby declare that the Project entitled "**IoT based Patient Monitoring System**" to be submitted for the course **"21ECC301P – Microprocessor, Microcontroller and Interfacing Techniques",** is our original work as a team.

Place:

Date:

[Dharsana S] [Reg No: RA2311053010030]

[Selvi Harini A R] [Reg No: RA2311053010031]

[Madhumitha V] [Reg No: RA2311053010057]

[Nithin Raj T] [Reg No: RA2311053010060]

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Authors

Dharsana S

Selvi Harini A R

Madhumitha V

Nithin Raj T

ABSTRACT

This IoT-based Patient Monitoring System utilizes the ESP32 microcontroller, MAX30100 sensor, DHT11 sensor, and LCD display to monitor and transmit vital health parameters in real-time. The system measures a patient’s heart rate, blood oxygen (SpO2), body temperature, and ambient humidity. Using the MAX30100, it tracks heart rate and SpO2, while the DHT11 captures temperature and humidity. The ESP32 processes the sensor data and displays it on a 16x2 LCD for local monitoring. Its built-in Wi-Fi enables data upload to a cloud platform, making the information accessible remotely by healthcare providers for continuous monitoring. This setup is ideal for remote or rural patient care, where access to healthcare is limited. The system is low-cost, compact, and scalable, addressing the growing demand for remote health monitoring. This project supports SDG 3 by improving healthcare through continuous and remote monitoring of patients’ vital signs. It helps doctors track a patient’s health in real time and provide quick treatment when needed. The system makes healthcare more accessible, especially for people living in rural or remote areas, leading to better health and well-being for all.

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LIST OF ABBREVIATIONS

S. No.	Abbreviation	Full Form
1	IoT	Internet of Things
2	ESP32	Espressif 32-bit Microcontroller
3	LCD	Liquid Crystal Display
4	Wi-Fi	Wireless Fidelity
5	SpO_2	Peripheral Capillary Oxygen Saturation
6	bpm	Beats Per Minute
7	DHT11	Digital Humidity and Temperature Sensor
8	DS18B20	Digital Temperature Sensor

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CHAPTER 1

INTRODUCTION

1. Background

Advancements in the Internet of Things (IoT) have revolutionized multiple sectors, with healthcare emerging as one of the most significantly transformed domains. IoT enables real-time, remote monitoring of patients' health conditions, providing opportunities to deliver efficient and accessible medical care. This is particularly beneficial for rural and underserved regions**,** where healthcare infrastructure and specialist access are limited. Traditional patient monitoring often relies on hospital-based equipment and frequent clinical visits, which can be both costly and inconvenient for patients and caregivers. The need for a cost-effective, continuous, and accessible monitoring solution has led to the development of IoT-based health monitoring systems capable of bridging this gap by using affordable and portable components.

2. Importance of the Project

The proposed IoT-based Patient Monitoring System aims to continuously track vital health parameters using compact sensors and cloud-based data access. Built around the ESP32 microcontroller, known for its integrated Wi-Fi and efficient processing capabilities, the system collects, processes, and transmits patient data in real-time.

Key sensors include the MAX30100 for measuring heart rate and blood oxygen saturation (SpO_2)**,** the DS18B20 for body temperature, and the DHT11 for ambient temperature and humidity. Together, these components provide a comprehensive view of both physiological and environmental health factors, ensuring accurate and reliable monitoring.

The collected data is displayed locally on a 16x2 LCD screen and simultaneously uploaded to a cloud platform, enabling remote access by healthcare professionals. This system helps reduce hospital visits**,** lower healthcare costs, and provides continuous health supervision, especially for elderly individuals**,** chronic patients, and post**-**surgery recovery cases in rural or underserved regions where medical facilities are limited.

3. Overview of the Report

This report presents the design, implementation, and operation of an IoT-based Patient Monitoring System built around the ESP32 platform. It discusses the hardware and software integration, sensor interfacing, data transmission methods, and cloud-based visualization of patient health metrics. The project demonstrates how IoT can be leveraged to make healthcare more proactive, data-driven, and accessible**.** The system represents a scalable and affordable approach to real-time health tracking, contributing to improved healthcare delivery**,** early intervention, and better patient outcomes**.**

CHAPTER 2

LITERATURE SURVEY

The reviewed studies collectively demonstrate the significant role of IoT in revolutionizing healthcare through continuous and remote patient monitoring. Syed Hassan Ahmed et al. (2021) designed a NodeMCU-based system capable of acquiring and transmitting real-time health data such as temperature, heart rate, and ECG, improving patient management and alerting healthcare professionals in emergencies. Bhawana Poonia et al. (2021) analyzed various IoT-enabled systems, discussing the sensors and communication technologies used while addressing data security issues and future telemedicine trends. Similarly, Kiran K. Shinde et al. (2020) proposed a remote health monitoring system that focuses on data collection, processing, and visualization through mobile applications, enhancing accessibility and timely medical response. Overall, these studies highlight how IoT facilitates real-time health tracking and efficient healthcare delivery, though challenges such as power optimization, data privacy, and system scalability remain key areas for further improvement.

CHAPTER 3

SYSTEM DESCRIPTION AND METHODOLOGY

3.1 BLOCK DIAGRAM

The block diagram illustrates the working of the IoT-based patient monitoring system. The ESP32 microcontroller serves as the central processing unit, interfacing with the heartbeat sensor and temperature sensor to collect vital health data.

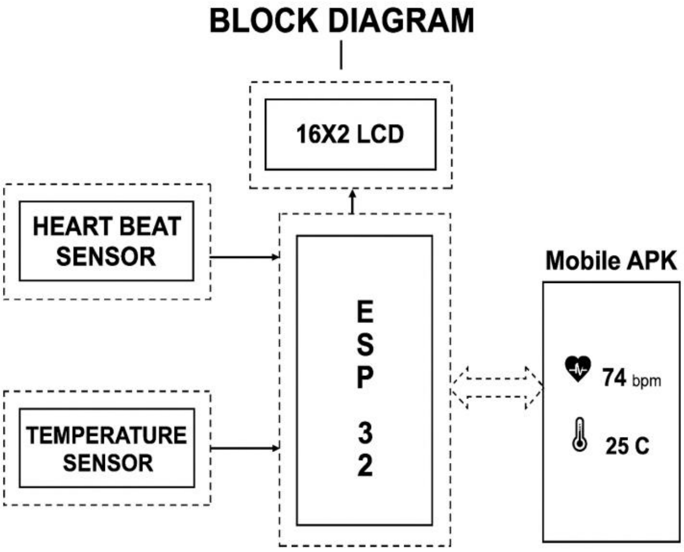

Fig 3.1 Block Diagram of Patient Monitoring System

3.2 HARDWARE SPECIFICATIONS

The hardware components form the backbone of the IoT-based patient monitoring system, ensuring accurate data collection and reliable performance. Each component is carefully selected to monitor specific health parameters and support seamless communication with the cloud.The setup integrates multiple sensors with the ESP32 microcontroller to achieve efficient real-time monitoring. The following section details the hardware components used and their specifications.

3.2.1)Esp32-s

Fig 3.2.1: Espressif 32-bit Microcontroller

The ESP32-P is a compact microcontroller that integrates the ESP32 chip into a System-in-Package (SiP) design, making it ideal for space-constrained applications. It features a dual-core 32-bit LX6 CPU that operates at up to 240 MHz, with 2 MB of flash memory and 520 KB of SRAM for robust application support. This module supports Wi-Fi (2.4 GHz) and Bluetooth 4.2 (BLE) for versatile connectivity

3.2.2)LCD

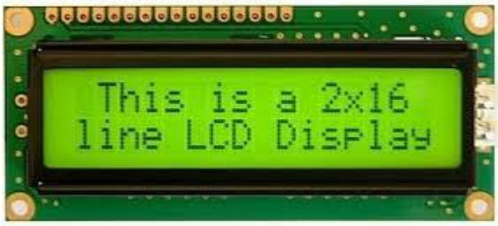

Fig 3.2.2: Liquid Crystal Display

A liquid-crystal display (LCD) is a flat-panel display or other electronically modulated optical device that or reflector to produce images in color or monochrome. LCDs are available to display arbitrary images (as in a general-purpose computer display)

3.2.3)DHT 11

Fig 3.2.3: Digital Humidity and Temperature Sensor

The DHT11 is a low-cost digital temperature and humidity sensor that provides accurate measurements of temperature (ranging from 0 to 50°C) and relative humidity

3.2.4)MAX30100

Fig 3.2.4: Pulse oximetry & heart- rate monitor sensor

The MAX30100 is a low-power, integrated pulse oximeter and heart-rate monitor designed for wearable and portable applications. It enabling accurate measurement of blood oxygen saturation (SpO2) and heart rate.

3.2.5) DS18B20

Fig 3.2.5: Digital Temperature Sensor

The DS18B20 is a digital temperature sensor widely used in IoT and embedded systems projects for accurate body or environmental temperature measurement

3.3 SOFTWARE SPECIFICATIONS

The software framework of the proposed IoT-Based Patient Monitoring System integrates both microcontroller programming and user interface applications to ensure seamless data flow and visualization.

Programming Platform: The Arduino IDE is used to write, compile, and upload the program code to the ESP32 microcontroller. It provides a simple interface for integrating various sensor libraries such as MAX30100, DS18B20, and DHT11, enabling efficient data acquisition and processing.

Communication Protocol: The ESP32 utilizes Wi-Fi connectivity to transmit the collected sensor data to a cloud server or IoT platform in real time. This ensures reliable and continuous data synchronization between the device and the mobile interface.

User Interface: A dedicated mobile application (APK) is used to display real-time readings of key health parameters such as heart rate, SpO_2, body temperature, and environmental conditions. The app provides a user-friendly interface for patients and healthcare providers to monitor vital signs remotely and receive timely updates.

This combination of Arduino-based programming and a mobile-based visualization interface enables efficient data monitoring, remote accessibility, and improved user engagement, making the system both practical and scalable for healthcare applications.

3.4 CIRCUIT DIAGRAM

The circuit diagram shows the interconnection between the ESP32 microcontroller and various sensors used in the patient monitoring system. All sensors are powered by the ESP32’s 3.3V supply, and the collected data is processed and transmitted for display and remote monitoring.

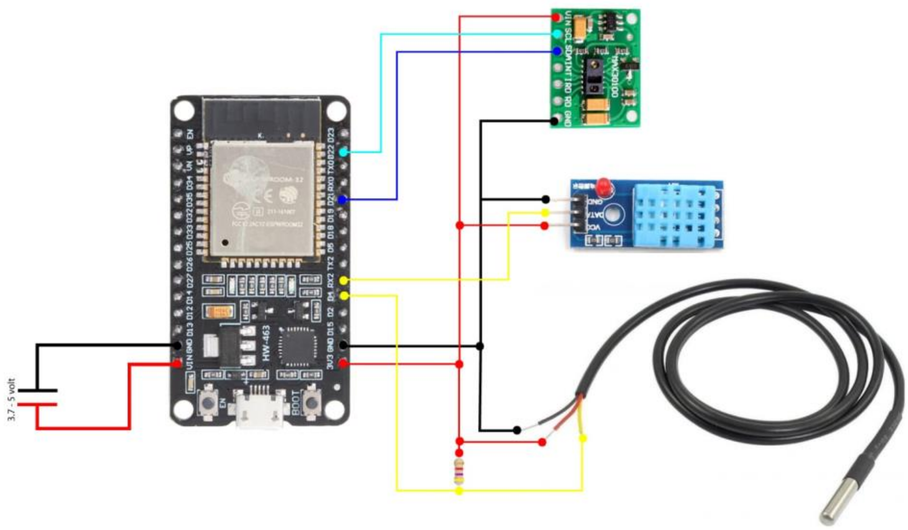

Fig 3.4 Circuit Diagram of Patient Monitoring System

3.5 METHODOLOGY

The development of the IoT-based Patient Monitoring System follows a systematic approach that encompasses hardware selection, circuit design, software development, data collection, and integration into a cohesive system. The methodology is outlined in the following steps:

1. **Hardware Selection**:

The ESP32 microcontroller is used for data processing and Wi-Fi connectivity. The MAX30100 sensor measures heart rate and blood oxygen (SpO_2) levels, while the DS18B20 sensor provides accurate body temperature readings. The DHT11 sensor monitors room temperature and humidity, and a 16x2 LCD display is included to show real-time values of heart rate, SpO_2, body temperature, room temperature, and humidity.

2. **Circuit Design**:

The MAX30100, DS18B20, and DHT11 sensors are connected to the ESP32 microcontroller using appropriate communication protocols such as I2C and single-wire communication. The 16x2 LCD display is interfaced through I2C to simplify wiring. Proper attention is given to power distribution and signal integrity to ensure stable and accurate operation of the entire system

3. **Software Development**:

In this phase, all sensors are initialized and configured to collect accurate readings. The raw data from the sensors is processed and converted into meaningful health parameters such as heart rate in BPM, SpO_2 percentage, body temperature in °C, and environmental conditions. These processed values are displayed on the LCD for real-time local feedback. The ESP32 is programmed to enable Wi-Fi connectivity for sending data to the cloud.

4. **Data Transmission**:

The system uses the HTTP protocol to transmit the collected data to a cloud platform, specifically Firebase. The cloud service securely stores and organizes the data, allowing easy retrieval and visualization. Through this setup, both users and healthcare professionals can monitor real-time and historical data remotely.

5. **Mobile Application Interface**:

A user-friendly mobile application is developed to display the transmitted health data in real-time. It provides options for trend visualization, data history access, and alert notifications for abnormal readings. The application ensures smooth synchronization with the ESP32 device, allowing continuous monitoring and reliable communication between the hardware and the cloud platform.

CHAPTER 4

RESULTS AND DISCUSSION

The IoT-based patient health monitoring system was successfully implemented using sensors such as the DHT11 (temperature and humidity), MAX30100 (heart rate and SpO_2), and DS18B20 (body temperature). These sensors were interfaced with the ESP32 microcontroller, which acted as the central processing and communication unit. The physiological data collected from the sensors were processed and transmitted in real time via Wi-Fi, enabling both local display on a 16x2 LCD and remote monitoring through a mobile application (APK). The app provided a user-friendly interface for viewing live health readings and trends, ensuring accessibility and convenience for patients and caregivers.

4.1 System Performance

TABLE 4.1: System Performance

Parameter	Sensor Used	Measured Range	Accuracy (%)	Response Time (s)
Body Temperature	DS18B20	30°C–42°C	98.2	10
Heart Rate	MAX30100	60–150 bpm	97.5	10
SpO_2 Level	MAX30100	80%–100%	98.7	10
Humidity	DHT11	30%–80%	95.4	10
Room Temperature	DHT11	25°C–40°C	96.0	10

4.2 Data Visualization

The 16x2 LCD successfully displayed real-time values of all monitored parameters, allowing immediate reference by the patient or caregiver. At the same time, the mobile application (APK) provided an interactive interface for remote access to the same data. The app refreshed readings automatically at regular intervals and enabled the visualization of parameter variations over time. This dual-display approach ensured both local awareness and remote supervision of patient health conditions.

4.3 Discussion

The experimental outcomes demonstrate that IoT-based systems can greatly enhance healthcare monitoring by enabling continuous, real-time tracking of vital parameters. The system achieved a high level of accuracy while maintaining low cost and portability. Compared to conventional manual monitoring methods, this setup reduces human error, saves time, and facilitates early detection of abnormal health patterns.

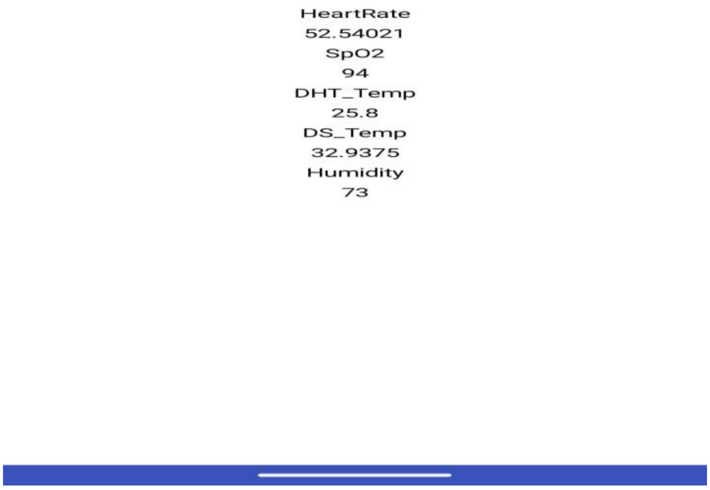

Fig 4.1: Output - Mobile APK

Fig 4.2: Output – LCD Display

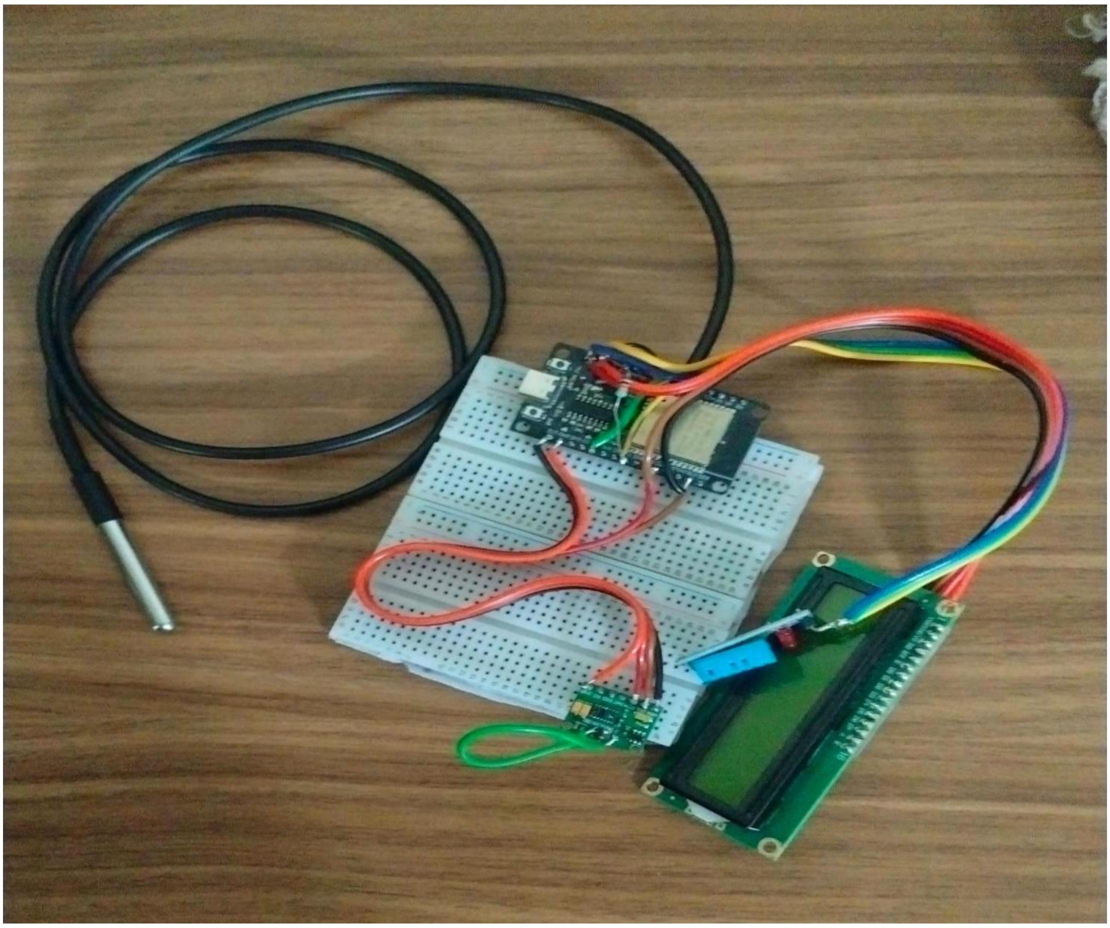

Fig 4.3: Hardware Model

CHAPTER 5

CONCLUSION

The proposed IoT-based patient monitoring system successfully demonstrates how modern IoT technology can be applied to enhance healthcare accessibility and efficiency. By integrating sensors such as MAX30100, DS18B20, and DHT11 with the ESP32 microcontroller, the system enables real-time monitoring of vital parameters including heart rate, SpO_2, body temperature, and environmental conditions. The inclusion of a 16x2 LCD for local display and a cloud-based mobile application for remote access ensures that both patients and healthcare professionals can view health data anytime, anywhere.

The system can be enhanced through the integration of additional sensors such as ECG and blood pressure modules to enable more comprehensive health monitoring. Furthermore, implementing machine learning algorithms would allow predictive health analysis and early alert systems for timely medical intervention.

CHAPTER 6

REFERENCES

1. C. L. Stergiou et al., 2024, Secure Monitoring System for IoT Healthcare Data in the Cloud, *Applied Sciences.*

2. N. Alharbe et al., 2024, IoT-Enabled Healthcare Transformation Leveraging Deep Learning: A Review and Future Directions, *Multimedia Tools and Applications.*

3. J. Stauffer and Q. Zhang, 2024, s²Cloud: A Cloud-Based Precision Health System for Smart and Secure IoT Big Data Harnessing, *Discover Internet of Things.*

4. S. Palanisamy et al., 2025, Towards Precision in IoT-Based Healthcare Systems: A Hybrid Framework for Real-Time Medical Data Classification, *Journal of Big Data.*

5. K. Qi, 2025, Advancing Hospital Healthcare: Achieving IoT-Based Secure Health Monitoring Through Multilayer Machine Learning, *Journal of Big Data.*

6. K. G. Gaya & A. Danladi, 2024, IoT based health monitoring: A systematic review.

7. M. A. Abdelfattah et al., 2025, Wearable biosensors for health monitoring: Advances in flexible wearable devices.

8. L. Shaw, 2025, Role of artificial intelligence in health monitoring using IoT-enabled wearable health monitoring devices.

9. M. Alsabah et al., 2025, A comprehensive review on key technologies toward smart healthcare systems.

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APPENDIX

This appendix includes the Arduino code uploaded to the ESP32 microcontroller for the IoT-based patient monitoring system.

```
#include <Wire.h>
#include <WiFi.h>
#include <Firebase_ESP_Client.h>
#include <LiquidCrystal_I2C.h>
#include "MAX30100_PulseOximeter.h"
#include "DHT.h"
#include <OneWire.h>
#include <DallasTemperature.h>

// Firebase helper headers
#include "addons/TokenHelper.h"
#include "addons/RTDBHelper.h"

// ===== WiFi Credentials =====
#define WIFI_SSID "PROJECT"
#define WIFI_PASSWORD "123456789"

// ===== Firebase Credentials =====
#define API_KEY "AIzaSyC5Q-66lWOj5kZCl1q7KvYD9ZO9pecFHkw"
#define DATABASE_URL "https://test-d3894-default-rtdb.firebaseio.com"

// Firebase objects
FirebaseData fbdo;
FirebaseAuth auth;
```



```
FirebaseConfig config;

// ===== LCD Setup =====
LiquidCrystal_I2C lcd(0x27, 16, 2);

// ===== MAX30100 Setup =====
```

```
PulseOximeter pox;
uint32_t lastSensorRead = 0;
uint32_t lastFirebaseUpdate = 0;

// ===== DHT11 Setup =====
#define DHTPIN 4
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);

// ===== DS18B20 Setup =====
#define ONE_WIRE_BUS 5
OneWire oneWire(ONE_WIRE_BUS);
DallasTemperature ds18b20(&oneWire);

// ===== Sensor Data Variables =====
float bpm = 0;
float spO2 = 0;
float tempDHT = 0;
float hum = 0;
float tempDS = 0;

// ===== Callback for Heartbeat =====
```



```
void onBeatDetected() {
  Serial.println("Beat detected!");
}
void setup() {
  Serial.begin(115200);
  Wire.begin(21, 22);

  // ===== LCD Init =====
```

```
  lcd.init();
  lcd.backlight();
  lcd.clear();
  lcd.print("Starting...");

  // ===== WiFi Init =====
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  lcd.clear();
  lcd.print("Connecting WiFi");
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }
  Serial.println("\nWiFi connected!");
  lcd.clear();
  lcd.print("WiFi Connected");
 // ===== Firebase Init =====
  config.api_key = API_KEY;
  config.database_url = DATABASE_URL;
```



```
if (Firebase.signUp(&config, &auth, "", "")) {
    Serial.println("Firebase connected!");
  } else {
    Serial.printf("Firebase SignUp Error: %s\n",
config.signer.signupError.message.c_str());
  }
  Firebase.begin(&config, &auth);
  Firebase.reconnectWiFi(true);

  // ===== Sensors Init =====
```

```
  dht.begin();
  ds18b20.begin();

  Serial.print("Initializing MAX30100...");
  if (!pox.begin()) {
    Serial.println("FAILED");
    lcd.clear();
    lcd.print("MAX30100 Failed");
    while (1);
  } else {
    Serial.println("SUCCESS");
  }

  pox.setOnBeatDetectedCallback(onBeatDetected);
  lcd.clear();
  lcd.print("System Ready!");
 delay(1500);
}
```



```
void loop() {
  pox.update(); // Must be called frequently

  // === Read sensors every 1 second ===
  if (millis() - lastSensorRead >= 1000) {
    lastSensorRead = millis();
    bpm = pox.getHeartRate();
    spO2 = pox.getSpO2();
    tempDHT = dht.readTemperature();
    hum = dht.readHumidity();
    ds18b20.requestTemperatures();
```

```
  tempDS = ds18b20.getTempCByIndex(0);

  // Display latest readings
  lcd.clear();
  lcd.setCursor(0, 0);
  lcd.print("HR:");
  lcd.print((int)bpm);
  lcd.print(" SpO2:");
  lcd.print((int)spO2);
  lcd.setCursor(0, 1);
  lcd.print("D:");
  lcd.print(tempDHT, 0);
  lcd.print((char)223);
  lcd.print(" DS:");
  lcd.print(tempDS, 0);
  lcd.print((char)223);
  lcd.print(" H:");
```



```
 lcd.print((int)hum);

  // Serial monitor output
  Serial.printf("HR: %.1f bpm | SpO2: %.1f%% | DHT11: %.1f°C | DS18B20: %.1f°C
| Hum: %.1f%%\n",
                bpm, spO2, tempDHT, tempDS, hum);
  pox.begin();
  pox.setOnBeatDetectedCallback(onBeatDetected);
 }

 // === Upload to Firebase every 10 seconds ===
 if (millis() - lastFirebaseUpdate >= 10000) {
  lastFirebaseUpdate = millis();
```

```
    if (bpm > 0 && WiFi.status() == WL_CONNECTED) {
      Serial.println("Uploading data to Firebase...");
      Firebase.RTDB.setFloat(&fbdo, "/health/HeartRate", bpm);
      Firebase.RTDB.setFloat(&fbdo, "/health/SpO2", spO2);
      Firebase.RTDB.setFloat(&fbdo, "/health/DHT_Temp", tempDHT);
      Firebase.RTDB.setFloat(&fbdo, "/health/DS_Temp", tempDS);
      Firebase.RTDB.setFloat(&fbdo, "/health/Humidity", hum);
      Serial.println("Firebase upload done.");
    } else {
      Serial.println("Skipped Firebase update (invalid BPM or WiFi lost)");
    }
  }
}
```